```
! pip install pypdf
```

```
Requirement already satisfied: pypdf in /usr/local/lib/python3.12/dist-packages (6.8.0)
```

```
! pip install chromadb
```

```
Requirement already satisfied: chromadb in /usr/local/lib/python3.12/dist-packages (1.5.5)
Requirement already satisfied: build>=1.0.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.4.0)
Requirement already satisfied: pydantic>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.12.3)
Requirement already satisfied: pydantic-settings>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.13.1)
Requirement already satisfied: pybase64>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.4.3)
Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.12/dist-packages (from uvicorn[standard]>=0.18.3->chro
Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.0.2)
Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.15.0)
Requirement already satisfied: onnxruntime>=1.14.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.24.3)
Requirement already satisfied: opentelemetry-api>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.40.0)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-grpc>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from c
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.40.0)
Requirement already satisfied: tokenizers>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.22.2)
Requirement already satisfied: pypika>=0.48.9 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.51.1)
Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.67.3)
Requirement already satisfied: overrides>=7.3.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (7.7.0)
Requirement already satisfied: importlib-resources in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.5.2)
Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.78.0)
Requirement already satisfied: bcrypt>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (5.0.0)
Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.24.1)
Requirement already satisfied: kubernetes>=28.1.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (35.0.0)
Requirement already satisfied: tenacity>=8.2.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (9.1.4)
Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.0.3)
Requirement already satisfied: mmh3>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (5.2.1)
Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.12/dist-packages (from chromadb) (3.11.7)
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (13.9.4)
Requirement already satisfied: jsonschema>=4.19.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.26.0)
Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.12/dist-packages (from build>=1.0.3->chromadb) (26.0)
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (2026.2.25)
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Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.27.0->chroma
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chromadb) (2
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chroma
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chromadb)
Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (1.17
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chro
Requirement already satisfied: websocket-client!=0.40.0,!=0.41.*,!=0.42.*,>=0.32.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (2.32.5
Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->ch
Requirement already satisfied: durationpy>=0.7 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb)
Requirement already satisfied: flatbuffers in /usr/local/lib/python3.12/dist-packages (from onnxruntime>=1.14.1->chromadb) (25
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from onnxruntime>=1.14.1->chromadb) (5.29.6
Requirement already satisfied: sympy in /usr/local/lib/python3.12/dist-packages (from onnxruntime>=1.14.1->chromadb) (1.14.0)
Requirement already satisfied: importlib-metadata<8.8.0,>=6.0 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-ap
Requirement already satisfied: googleapis-common-protos~=1.57 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-ex
Requirement already satisfied: opentelemetry-exporter-otlp-proto-common==1.40.0 in /usr/local/lib/python3.12/dist-packages (fro
Requirement already satisfied: opentelemetry-proto==1.40.0 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-expo
Requirement already satisfied: opentelemetry-semantic-conventions==0.61b0 in /usr/local/lib/python3.12/dist-packages (from oper
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.0->chromadb
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.0->chromadb)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.0->chroma
```

```
! pip install langchain-community
```

```
Requirement already satisfied: langchain-community in /usr/local/lib/python3.12/dist-packages (0.4.1)
Requirement already satisfied: langchain-core<2.0.0,>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from langchain-community
Requirement already satisfied: langchain-classic<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langchain-commun
Requirement already satisfied: SQLAlchemy<3.0.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (2
Requirement already satisfied: requests<3.0.0,>=2.32.5 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (2.
Requirement already satisfied: PyYAML<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (6.0.3
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (3.13
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-commun
Requirement already satisfied: dataclasses-json<0.7.0,>=0.6.7 in /usr/local/lib/python3.12/dist-packages (from langchain-communi
Requirement already satisfied: pydantic-settings<3.0.0,>=2.10.1 in /usr/local/lib/python3.12/dist-packages (from langchain-commu
Requirement already satisfied: langsmith<1.0.0,>=0.1.125 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (0.
Requirement already satisfied: numpy>=1.26.2 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (2.0.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->l
```

```
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchai
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain-c
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langcha
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langc
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchai
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langcha
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.7.
Requirement already satisfied: typing-inspect<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.7.0,
Requirement already satisfied: langchain-text-splitters<2.0.0,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain-classic<2.0.0,>
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=
Requirement already satisfied: packaging>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.0.1->
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->lang
Requirement already satisfied: orjson>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langch
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.1
Requirement already satisfied: xxhash>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langcha
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->lan
Requirement already satisfied: python-dotenv>=0.21.0 in /usr/local/lib/python3.12/dist-packages (from pydantic-settings<3.0.0,>=
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from pydantic-settings<3.0.0
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->langchain-
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->lang
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->lang
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from SQLAlchemy<3.0.0,>=1.4.0->langchain-
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->langsmith<1.0.0,>=0.1.12
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->langsmith<1.0.0,
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->langs
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->langc
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->l
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->la
Requirement already satisfied: mypy-extensions>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from typing-inspect<1,>=0.4.0-
```

```
import openai
import os
import langchain
```

```
from google.colab import userdata
os.environ['OPENAI_API_KEY'] = userdata.get('OPENAI_API_KEY')
```

```
from google.colab import drive
drive.mount('/content/drive', force_remount = True)
```

```
Mounted at /content/drive
```

```
!pip install -U -qq llama-index openai
```

```
from llama_index.core import SimpleDirectoryReader

# Let us take input from a directory
reader = SimpleDirectoryReader(input_dir="/content/drive/MyDrive/MS_Experiment_Results")
```

```
from llama_index.llms.openai import OpenAI
from llama_index.core.llms import ChatMessage
import os
import openai
import pandas as pd
```

```
# Use the load_data() method to read the files from the directory
documents = reader.load_data()
# number of files
print(f"Loaded {len(documents)} docs")
```

```
Loaded 475 docs
```

```
from llama_index.core import VectorStoreIndex, SimpleDirectoryReader
# Load documents and build index
index = VectorStoreIndex.from_documents(documents)
```

```
from llama_index.core.node_parser import SimpleNodeParser
from llama_index.core import VectorStoreIndex
from IPython.display import display, HTML
```

```
# create parser and parse document into nodes
parser = SimpleNodeParser.from_defaults()
nodes = parser.get_nodes_from_documents(documents)

# # build index
index = VectorStoreIndex(nodes)

# Construct Query Engine
query_engine = index.as_query_engine()
```

```
response = query_engine.query("Give me EDA details for Batsmen?")
```

```
type(response)
```

```
llama_index.core.base.response.schema.Response
```

```
## Query response function
def query_response(user_input):
   response = query_engine.query(user_input)
   file_name = response.source_nodes[0].node.metadata['file_name'] + " page nos " + response.source_nodes[0].node.metadata['pag
   final_response = response.response + '\n Check further at ' + file_name
   return final_response
```

```
def initialize_conv():
  print('Feel free to ask Questions regarding IPL document. Press exit once you are done')
  while True:
    user_input = input()
    # Type 'exit' to exit conversation
    if user_input.lower() == 'exit':
      print('Exiting the program... bye')
      break
    else:
      response = query_response(user_input)
      display(HTML(f'<p style="font-size:20px">{response}</p>'))
```

```
initialize_conv()
```

```
Feel free to ask Questions regarding IPL document. Press exit once you are done
Accuracy of random forest for different models?
```

The Accuracy of the Random Forest Classifier (Base) is 0.86, while the hyperparameter-tuned Random Forest Classifier achieved an Accuracy of 0.85. Check further at MS_Implementation_Wks_Price_Prediction [CNN + Gen AI + ML].pdf page nos 84,89

```
CNN use?
```

The CNN's poor performance, especially with reduced data, suggests that this architecture may not be optimally suited for the problem at hand. CNNs typically require more data to learn meaningful features, and their strength lies in capturing spatial or temporal patterns, which may not be prominent in the preprocessed tabular dataset used in this scenario. Check further at MS_Implementation_AllRounders_Price_Prediction.ipynb [AI+ML+CNN].pdf page nos 99,87

```
Bagging & Boosting?
```

Bagging and Boosting classifiers are considered more robust and capable of handling noise better in real-world scenarios, even if they demonstrate perfect scores on a specific dataset. The Bagging Classifier with optimized parameters achieved perfect scores on the test set and high scores on the validation set, indicating its effectiveness in classifying the 'Price Category' for the dataset. The Boosting Classifier is being subjected to hyperparameter tuning to optimize its performance and prevent overfitting, following a similar approach to the

```
questions = ['What is the approach used for prediction?','What is the maximum accuacy of prediction of wicketkeepers?','What is
```

```
def testing_pipeline(questions):
  test_feedback  = []
  for i in questions:
    print(i)
    print(query_response(i))
    print('\n Please provide your feedback on the response provided by the bot')
    user_input = input()
    page = query_response(i).split()[-1]
    test_feedback.append((i,query_response(i),page,user_input))
```

```
  feedback_df = pd.DataFrame(test_feedback, columns =['Question', 'Response', 'Page','Good or Bad'])
  return feedback_df
```

```
testing_pipeline(questions)
```

```
What is the approach used for prediction?
The approach used for prediction involves blending traditional Ensemble Machine Learning with Agentic Generative AI to predict a
 Check further at MS Project Final Implementation Framework.pdf page nos 1,16

 Please provide your feedback on the response provided by the bot
Good
What is the maximum accuacy of prediction of wicketkeepers?
The maximum accuracy of prediction for wicketkeepers is 86.00%.
 Check further at MS Project Final Implementation Framework.pdf page nos 15,7

 Please provide your feedback on the response provided by the bot
Good
What is the minimum accuracy of combined?
The minimum accuracy of the combined models is $75.00\%$.
 Check further at MS Project Final Implementation Framework.pdf page nos 7,6

 Please provide your feedback on the response provided by the bot
Ok
```

	Question	Response	Page	Good or Bad
0	What is the approach used for prediction?	The approach used for prediction involves blen...	1,16	Good
1	What is the maximum accuacy of prediction of w...	The maximum accuracy of prediction for wicketk...	15,7	Good
2	What is the minimum accuracy of combined?	The minimum accuracy of the combined models is...	7,6	Ok

```
# 1. Use the updated import path (recommended for LangChain 0.1.0+)
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_core.documents import Document as LCDocument # Alias to avoid name collision with llama_index.core.Document

# 2. Initialize the splitter
text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=500,
    chunk_overlap=50,
    length_function=len,
    is_separator_regex=False,
)

# 3. Split the documents
# Convert llama_index.core.Document objects to LangChain Document objects
langchain_documents = []
for doc in documents:
    langchain_documents.append(LCDocument(page_content=doc.text, metadata=doc.metadata))

chunks = text_splitter.split_documents(langchain_documents)

# 4. View the result
chunks
```

```
[Document(metadata={'page_label': '1', 'file_name': 'MS Project Final Implementation Framework.pdf', 'file_path':
'/content/drive/MyDrive/MS_Experiment_Results/MS Project Final Implementation Framework.pdf', 'file_type': 'application/pdf',
'file_size': 333908, 'creation_date': '2026-03-11', 'last_modified_date': '2026-03-11'}, page_content='Hybrid AI-Driven
Valuation for IPL Auctions \nSubject: Integration of Predictive ML and Agentic Generative AI \n \n1. Executive Summary \nThis
project architectures a high-fidelity valuation system for the Indian Premier League \n(IPL) auction. By blending traditional
Ensemble Machine Learning with Agentic \nGenerative AI, the system predicts a player’s "Fair Market Value" and provides a
real-time \nstrategic "Scouting Report" for bidding wars. \n \nLayer 1: The Analytical Base (ML Algos)'),
 Document(metadata={'page_label': '1', 'file_name': 'MS Project Final Implementation Framework.pdf', 'file_path':
'/content/drive/MyDrive/MS_Experiment_Results/MS Project Final Implementation Framework.pdf', 'file_type': 'application/pdf',
'file_size': 333908, 'creation_date': '2026-03-11', 'last_modified_date': '2026-03-11'}, page_content='Layer 1: The Analytical
Base (ML Algos) \nWe utilize a suite of regression models to establish a baseline market value ($V_s$) based \non historical
performance data (2008–2025). \n• Linear/Polynomial Regression: Identifying trends between strike rates and price. \n• Support
Vector Regression (SVR): Handling high-value outliers and non-linear \nperformance jumps. \n• Random Forest & Decision Trees:
Capturing complex feature interactions (e.g., the \n"All-rounder" premium).'),
 Document(metadata={'page_label': '1', 'file_name': 'MS Project Final Implementation Framework.pdf', 'file_path':
'/content/drive/MyDrive/MS_Experiment_Results/MS Project Final Implementation Framework.pdf', 'file_type': 'application/pdf',
'file_size': 333908, 'creation_date': '2026-03-11', 'last_modified_date': '2026-03-11'}, page_content='"All-rounder" premium).
\nLayer 2 : Predictive Engine: Ensemble Learning \nTo handle the volatility of auction prices, the model employs two primary
ensemble \ntechniques to stabilize and refine predictions. \n2.1 Bagging (Random Forest) \n• Role: Establishes a stable "Price
Floor. " \n• Logic: By averaging multiple decision trees, it reduces variance and prevents the \nmodel from overreacting to
anomaly bids (outliers). \n2.2 Boosting (XGBoost / Gradient Boosting)'),
 Document(metadata={'page_label': '1', 'file_name': 'MS Project Final Implementation Framework.pdf', 'file_path':
'/content/drive/MyDrive/MS_Experiment_Results/MS Project Final Implementation Framework.pdf', 'file_type': 'application/pdf',
'file_size': 333908, 'creation_date': '2026-03-11', 'last_modified_date': '2026-03-11'}, page_content='2.2 Boosting (XGBoost /
Gradient Boosting) \n• Role: Captures the "Elite Premium. " \n• Logic: Sequentially corrects errors from previous trees to
```

```
identify the price spikes \nassociated with high-demand marquee players.'),
 Document(metadata={'page_label': '2', 'file_name': 'MS Project Final Implementation Framework.pdf', 'file_path':
'/content/drive/MyDrive/MS_Experiment_Results/MS Project Final Implementation Framework.pdf', 'file_type': 'application/pdf',
'file_size': 333908, 'creation_date': '2026-03-11', 'last_modified_date': '2026-03-11'}, page_content='Layer 3. Generative AI
& Data Augmentation [Testing & Validation] \nTo overcome limited historical data, the framework uses Generative AI to simulate
market \nconditions. \n• Variational Autoencoders (VAEs): Generates thousands of synthetic player profiles \nto stress-test
the model. \n• Diffusion Models: Simulates bidding "heatmaps" to predict how prices escalate \nduring specific rounds of the
auction. \n \nLayer 4. Strategic Layer: Agentic AI Scout'),
 Document(metadata={'page_label': '2', 'file_name': 'MS Project Final Implementation Framework.pdf', 'file_path':
'/content/drive/MyDrive/MS_Experiment_Results/MS Project Final Implementation Framework.pdf', 'file_type': 'application/pdf',
'file_size': 333908, 'creation_date': '2026-03-11', 'last_modified_date': '2026-03-11'}, page_content='Layer 4. Strategic
Layer: Agentic AI Scout \nThe Agentic Scout serves as the system’s reasoning engine. It utilizes a ReAct (Reasoning \n+
Acting) loop to orchestrate tools. \n4.1 The Reasoning Workflow \n1. Query: Receives a player name and team budget.  \n2. Act:
Calls the predict_market_value tool (Random Forest/XGBoost). \n3. Observe: Analyzes current squad gaps and market scarcity.
\n4. Recommend: Outputs a final "Bid Ceiling" and a pivot strategy. \n6. Conclusion'),
 Document(metadata={'page_label': '2', 'file_name': 'MS Project Final Implementation Framework.pdf', 'file_path':
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'file_size': 333908, 'creation_date': '2026-03-11', 'last_modified_date': '2026-03-11'}, page_content='6. Conclusion \nThe
hybrid approach ensures that the final valuation is both statistically grounded and \nstrategically adaptive. By combining the
stability of Bagging, the precision of Boosting, and \nthe reasoning of Agentic AI, the model provides a competitive edge in
high-stakes auction \nenvironments.'),
 Document(metadata={'page_label': '3', 'file_name': 'MS Project Final Implementation Framework.pdf', 'file_path':
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'file_size': 333908, 'creation_date': '2026-03-11', 'last_modified_date': '2026-03-11'}, page_content='Research Monograph: AI-
Driven Multi-\nDimensional Cricket Player Auction \nValuation \n \n1. Executive Summary \nThis comprehensive technical
research report details the end-to-end development of a \nsophisticated appraisal engine designed to quantify and predict the
market value of \nprofessional cricket players. By integrating advanced Exploratory Data Analysis \n(EDA), Machine Learning
(ML), and Deep Learning (Convolutional Neural'),
 Document(metadata={'page_label': '3', 'file_name': 'MS Project Final Implementation Framework.pdf', 'file_path':
'/content/drive/MyDrive/MS_Experiment_Results/MS Project Final Implementation Framework.pdf', 'file_type': 'application/pdf',
'file_size': 333908, 'creation_date': '2026-03-11', 'last_modified_date': '2026-03-11'}, page_content='Networks - CNN), the
project transitions cricket analytics from descriptive statistics to \npredictive financial modeling. \nThe study bifurcates
the player population into four tactical categories: Batsmen, \nBowlers, All-Rounders, and Wicketkeepers. The primary
```

```
chunks[50].page_content
```

```
'to new, unseen data rather than just memorizing labels. \n \n7. Strategic Recommendations & Conclusion \n7.1 Technical Summary
\n1. Architecture: For tabular sports statistics, Ensemble Methods (Random \nForest/Bagging) are superior to Deep Learning (CNN
s). They offer higher accuracy \nwith significantly less computational cost. \n2. Scaling: Future versions should incorporate
```

```
from langchain_community.embeddings import HuggingFaceEmbeddings
embeddings=HuggingFaceEmbeddings()
```

```
/tmp/ipykernel_23782/1099809318.py:2: LangChainDeprecationWarning: The class `HuggingFaceEmbeddings` was deprecated in LangChain
  embeddings=HuggingFaceEmbeddings()
/tmp/ipykernel_23782/1099809318.py:2: LangChainDeprecationWarning: Default values for HuggingFaceEmbeddings.model_name were depr
  embeddings=HuggingFaceEmbeddings()
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Loading weights: 100%                                                    199/199 [00:00<00:00, 371.62it/s, Materializing param=pooler.dense.weight]
MPNetModel LOAD REPORT from: sentence-transformers/all-mpnet-base-v2
Key                      | Status     |  |
-------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
```

```
from langchain_community.vectorstores import Chroma
```

```
vector_store=Chroma.from_documents(documents=chunks,embedding=embeddings)

retriever=vector_store.as_retriever(search_kwargs={'k':5})
retriever
```

```
VectorStoreRetriever(tags=['Chroma', 'HuggingFaceEmbeddings'], vectorstore=<langchain_community.vectorstores.chroma.Chroma
object at 0x7a04729166c0>, search_kwargs={'k': 5})
```

```
!pip install langchain-openai
from langchain_openai import ChatOpenAI
```

```
llm=ChatOpenAI(model_name='gpt-3.5-turbo',temperature=0.0)
llm
```

```
Requirement already satisfied: langchain-openai in /usr/local/lib/python3.12/dist-packages (1.1.11)
Requirement already satisfied: langchain-core<2.0.0,>=1.2.18 in /usr/local/lib/python3.12/dist-packages (from langchain-openai)
Requirement already satisfied: openai<3.0.0,>=2.26.0 in /usr/local/lib/python3.12/dist-packages (from langchain-openai) (2.26.0)
Requirement already satisfied: tiktoken<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-openai) (0.12.0
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=
Requirement already satisfied: packaging>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.18-
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.26.0->langchain
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.26.0->langchai
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.26.0->langchai
Requirement already satisfied: jiter<1,>=0.10.0 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.26.0->langchai
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.26.0->langchain-openai)
Requirement already satisfied: tqdm>4 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.26.0->langchain-openai)
Requirement already satisfied: regex>=2022.1.18 in /usr/local/lib/python3.12/dist-packages (from tiktoken<1.0.0,>=0.7.0->langcha
Requirement already satisfied: requests>=2.26.0 in /usr/local/lib/python3.12/dist-packages (from tiktoken<1.0.0,>=0.7.0->langcha
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5,>=3.5.0->openai<3.0.0,>=2.26.0
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->openai<3.0.0,>=2.26.0-
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->openai<3.0.0,>=2
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->opena
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->langc
Requirement already satisfied: orjson>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langcha
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.4
Requirement already satisfied: xxhash>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langchai
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->lang
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->l
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->la
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4-
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.26.0->tikto
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.26.0->tiktoken<1.
ChatOpenAI(profile={'max_input_tokens': 16385, 'max_output_tokens': 4096, 'text_inputs': True, 'image_inputs': False,
'audio_inputs': False, 'video_inputs': False, 'text_outputs': True, 'image_outputs': False, 'audio_outputs': False,
'video_outputs': False, 'reasoning_output': False, 'tool_calling': False, 'structured_output': False, 'image_url_inputs':
False, 'pdf_inputs': False, 'pdf_tool_message': False, 'image_tool_message': False, 'tool_choice': True}, client=
<openai.resources.chat.completions.completions.Completions object at 0x7a0470e02ae0>, async_client=
<openai.resources.chat.completions.completions.AsyncCompletions object at 0x7a0467345e80>, root_client=<openai.OpenAI object at
0x7a0470e48bf0>, root_async_client=<openai.AsyncOpenAI object at 0x7a04673457c0>, temperature=0.0, model_kwargs={},
openai_api_key=SecretStr('**********'), stream_usage=True)
```

```
!pip install langchain_experimental
import pandas as pd
from langchain_openai import ChatOpenAI
from langchain_experimental.agents import create_pandas_dataframe_agent
import os # Added import for os

# 1. Load the Excel File
# Use 'openpyxl' engine for .xlsx files
file_path = "/content/drive/MyDrive/MS_Experiment_Results/Combined_Cricket_Data.csv"
df = pd.read_csv(file_path)

# 2. Initialize the LLM (Large Language Model)
# Temperature is set to 0 for consistent, factual data analysis
llm = ChatOpenAI(model="gpt-3.5-turbo", temperature=0, openai_api_key=os.environ['OPENAI_API_KEY']) # Using os.environ

# 3. Create the Agentic AI
# allow_dangerous_code=True is required as the agent must execute
# python code to query the dataframe.
agent_executor = create_pandas_dataframe_agent(
    llm,
    df,
    agent_type="openai-tools",
    verbose=True,
    allow_dangerous_code=True
)

# 4. Query the Agent
query = "Which player has the highest strike rate among Bowlers from India?"
response = agent_executor.invoke({"input": query})

print(f"Final Answer: {response['output']}")
```

```
Requirement already satisfied: langchain_experimental in /usr/local/lib/python3.12/dist-packages (0.4.1)
Requirement already satisfied: langchain-core<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langchain_experime
```

```
Requirement already satisfied: langchain-community<1.0.0,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain_exp
Requirement already satisfied: langchain-classic<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langchain-commu
Requirement already satisfied: SQLAlchemy<3.0.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community<1
Requirement already satisfied: requests<3.0.0,>=2.32.5 in /usr/local/lib/python3.12/dist-packages (from langchain-community<1.0
Requirement already satisfied: PyYAML<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community<1.0.0
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.12/dist-packages (from langchain-community<1.0.0
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-commu
Requirement already satisfied: dataclasses-json<0.7.0,>=0.6.7 in /usr/local/lib/python3.12/dist-packages (from langchain-commun
Requirement already satisfied: pydantic-settings<3.0.0,>=2.10.1 in /usr/local/lib/python3.12/dist-packages (from langchain-comm
Requirement already satisfied: langsmith<1.0.0,>=0.1.125 in /usr/local/lib/python3.12/dist-packages (from langchain-community<1
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community<1.0
Requirement already satisfied: numpy>=1.26.2 in /usr/local/lib/python3.12/dist-packages (from langchain-community<1.0.0,>=0.4.0
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,
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Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<
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Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->lang
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langcha
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langch
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.7
Requirement already satisfied: typing-inspect<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.7.0
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->lang
Requirement already satisfied: langchain-text-splitters<2.0.0,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from langcha
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->lan
Requirement already satisfied: orjson>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langc
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1
Requirement already satisfied: xxhash>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langch
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->la
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4-
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4
Requirement already satisfied: python-dotenv>=0.21.0 in /usr/local/lib/python3.12/dist-packages (from pydantic-settings<3.0.0,
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->langchain
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->lan
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->lan
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from SQLAlchemy<3.0.0,>=1.4.0->langchain
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->langsmith<1.0.0,>=0.1.1
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->langsmith<1.0.0
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->lang
Requirement already satisfied: mypy-extensions>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from typing-inspect<1,>=0.4.0


> Entering new AgentExecutor chain...

Invoking: `python_repl_ast` with `{'query': "df[df['Role'].str.contains('Bowler') & (df['Country'] == 'India')].sort_values('Ba


Suyash SharmaThe player with the highest strike rate among Bowlers from India is Suyash Sharma.
```

```
import pandas as pd
import numpy as np

def query_response(user_input):
    """
    Logic for querying the pandas dataframe agent.
    """
    try:
        response = agent_executor.invoke({"input": user_input})
        return response['output']
    except Exception as e:
        return f"Error retrieving data from CSV: {str(e)}"

def initialize_conv():
    """
    Standard CLI loop for sequential questions regarding Player Valuation.
    """
    print('--- Cricket Player Valuation Assistant ---')
    print('Ask questions regarding your loaded CSV data.')
    print('Type "exit" to close the session.')

    while True:
        user_input = input("\nYou: ")

        if user_input.lower() == 'exit':
```

```
            print('Exiting the program... bye')
            break
        else:
            # Generate the response
            final_answer = query_response(user_input)

            # Print the formatted output
            print("-" * 50)
            print(final_answer)
            print("-" * 50)

# Start the conversation
if __name__ == "__main__":
    initialize_conv()
```

```
--- Cricket Player Valuation Assistant ---
Ask questions regarding your loaded CSV data.
Type "exit" to close the session.

You: What is the auction price of Virat Kohli?


> Entering new AgentExecutor chain...

Invoking: `python_repl_ast` with `{'query': "df[df['Player']=='Virat Kohli']['Last Auction Price Sold (in Crs)'].values[0]"}`


21.0The auction price of Virat Kohli is 21.0 Crs.

> Finished chain.
--------------------------------------------------
The auction price of Virat Kohli is 21.0 Crs.
--------------------------------------------------

You: The runs scored by Hardik pandya?


> Entering new AgentExecutor chain...

Invoking: `python_repl_ast` with `{'query': "df[df['Player'] == 'Hardik Pandya']['Runs Scored'].values[0]"}`


2757.0Hardik Pandya has scored 2757 runs.

> Finished chain.
--------------------------------------------------
Hardik Pandya has scored 2757 runs.
--------------------------------------------------

You: you have 12 crs select me 5 batsmen?


> Entering new AgentExecutor chain...

Invoking: `python_repl_ast` with `{'query': "df[df['Last Auction Price Sold (in Crs)'] <= 12].nlargest(5, 'Batting Average')[[


            Player  Batting Average  Last Auction Price Sold (in Crs)
1837    Player_310           134.81                              0.07
1463  Aaron George           105.00                              0.00
1550     Player_23            90.24                              2.70
1834    Player_307            63.51                              0.38
1828    Player_301            63.43                              0.17Here are 5 batsmen within your budget of 12 Crs based on 

1. Player_310 - Batting Average: 134.81, Last Auction Price Sold: 0.07 Crs
2. Aaron George - Batting Average: 105.00, Last Auction Price Sold: 0.00 Crs
3. Player_23 - Batting Average: 90.24, Last Auction Price Sold: 2.70 Crs
4. Player_307 - Batting Average: 63.51, Last Auction Price Sold: 0.38 Crs
5. Player_301 - Batting Average: 63.43, Last Auction Price Sold: 0.17 Crs

> Finished chain.
--------------------------------------------------
```